DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Session: 2025-26

PROFESSIONAL ELECTIVE II - DATA WAREHOUSING AND MINING

Case Study on

CUSTOMER SEGMENTATION USING K-MEANS CLUSTERING IN WEKA

Submitted by

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1. Introduction

In today's data-driven world, organizations generate vast amounts of data through transactions, customer interactions, and digital platforms. Managing and extracting useful information from this data is essential for strategic decision-making.

Data Warehousing involves collecting and storing large volumes of structured data from multiple sources into a centralized repository. Once stored, this data is analysed using Data Mining techniques to discover patterns, trends, and relationships.

One such important technique is clustering, which groups similar data points without predefined labels. In this project, clustering is used to segment customers based on purchasing behaviour.

This case study utilizes WEKA, a popular open-source data mining tool, to implement the K-Means clustering algorithm on a customer dataset.

2. Objectives

- To understand the concept of data preprocessing and transformation
- To apply unsupervised learning techniques (clustering)
- To implement K-Means algorithm using WEKA
- To visualize and interpret customer segments
- To analyze business insights from clustering results

3. Background

Before applying data mining, data is organized using a data warehouse.

Key Concepts:

- ETL Process (Extract, Transform, Load)
  - Extract data from sources
  - Transform into usable format
  - Load into warehouse
- Data Cleaning
- Data Integration
- Data Transformation

Here, our CSV dataset is our Source data and Data Preprocessing is done in WEKA (Transformation step).

4. Dataset Description

The dataset used for this Case Study was sourced from **Kaggle**, named "**Mall Customer Segmentation Dataset**".

This dataset contains 1,000 customer records suitable for segmentation, clustering, and behavioral analysis tasks. It includes demographic attributes, income information, and a spending behavior score designed for analytical modeling.

Data Characteristics

- Age distribution is concentrated within economically active ranges (primarily 22–49).
- Income values follow realistic career progression patterns.
- Spending Score reflects behavioral variation across age and income groups.
- Each feature column (excluding CustomerID) contains a small number of missing values (3–6 per column) to support data cleaning and preprocessing exercises.
- Value ranges are controlled to maintain economic plausibility.

Attributes:

Attribute	Type	Description
CustomerID	Numeric	Unique identifier
Gender	Nominal	Male/Female
Age	Numeric	Age of customer
Annual Income (k$)	Numeric	Income
Spending Score (1–100)	Numeric	Spending behavior

The dataset was chosen because it is:

- Helpful in applying Missing value handling techniques
- Suitable for clustering techniques
- Easy to preprocess and visualize
- Effective for demonstrating meaningful customer segmentation
- Structured for practical analytics workflows and educational projects involving customer behaviour analysis.

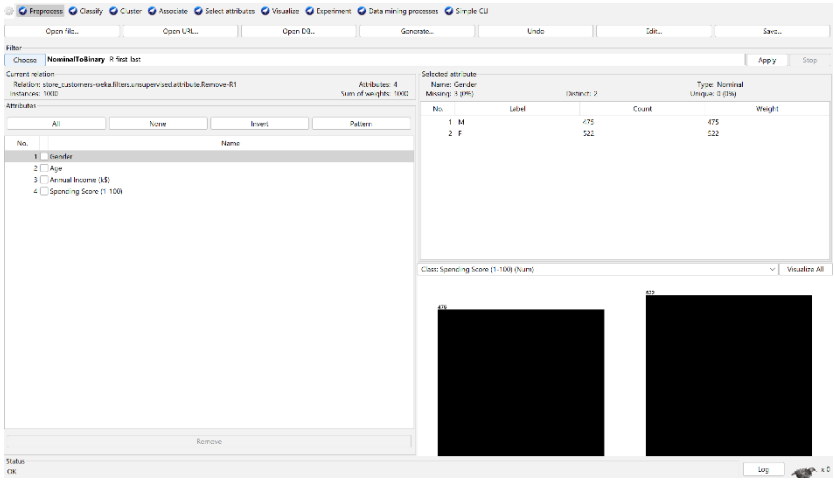
5. Data pre-processing

Data preprocessing is essential for improving clustering accuracy.

Steps Performed:

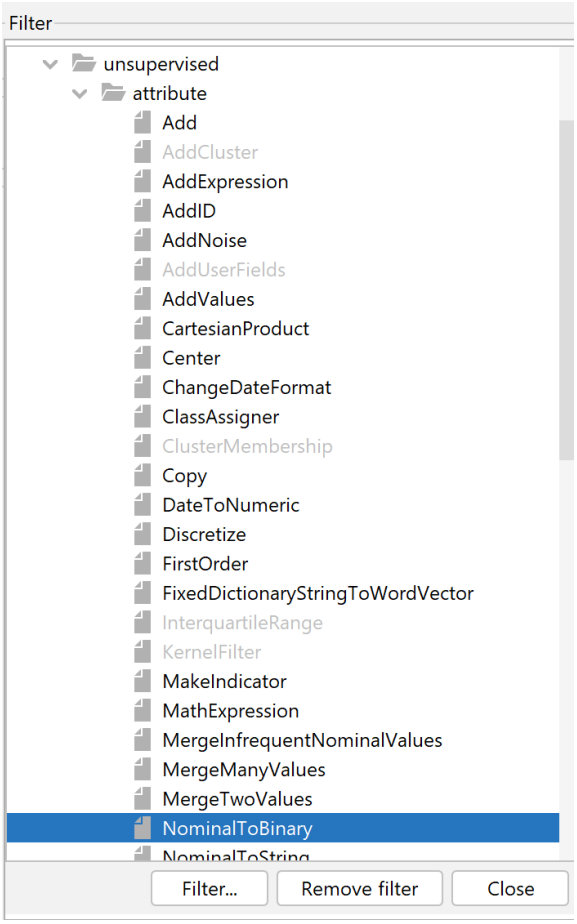
5.1 Removing Irrelevant Attribute

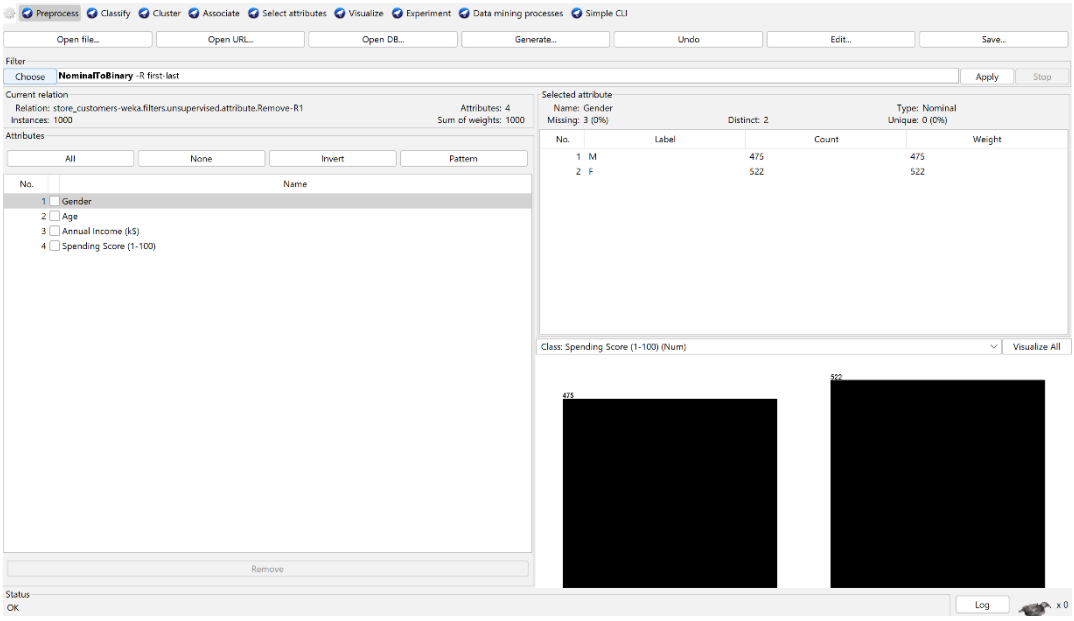
- Removed CustomerID
- Reason: Does not contribute to clustering



5.2 Converting Categorical Data

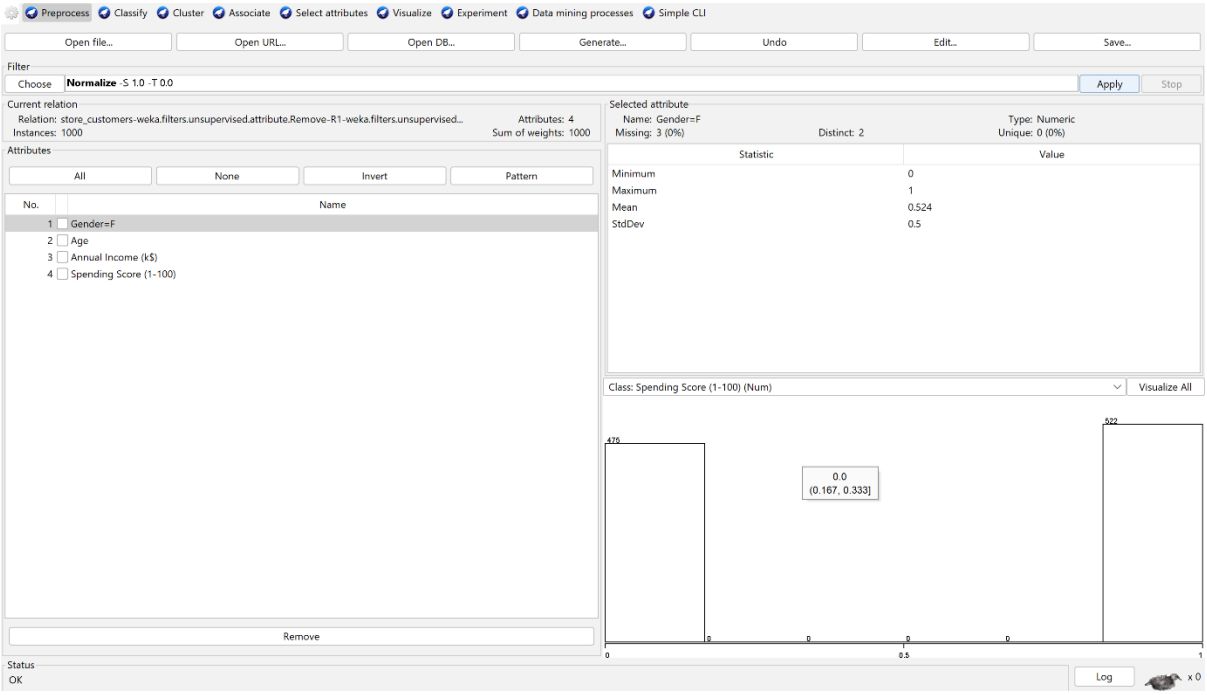
- Converted Gender → Numeric
- Used: NominalToBinary filter





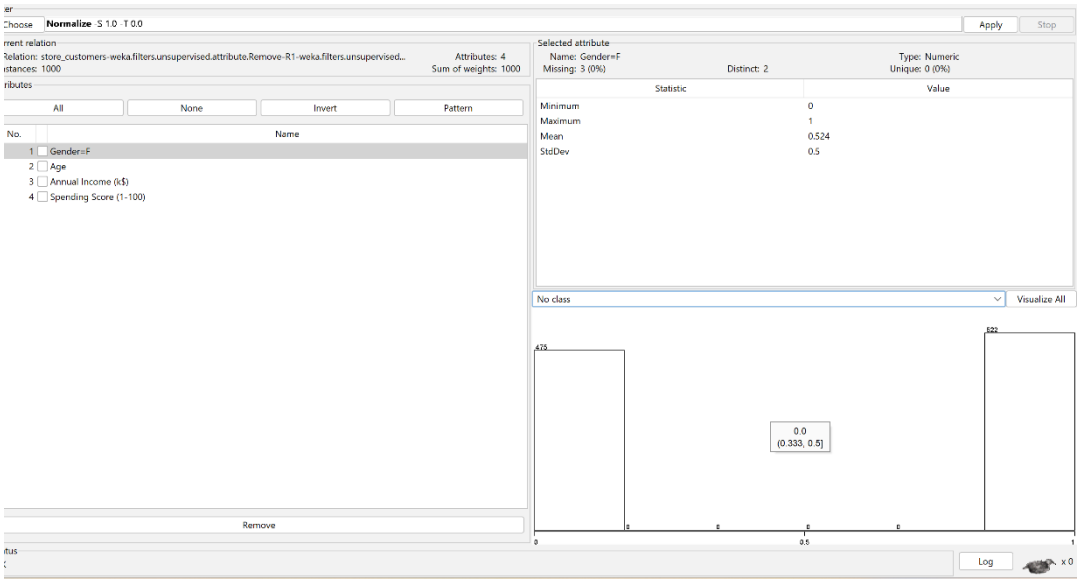
5.3 Normalization

- Applied **Normalize filter**
- Scaled values between 0 and 1 for Gender
- Important because clustering uses distance



4.4 Removing Class Label

- Set class attribute to **None**
- Reason: Clustering is unsupervised learning



5. Algorithm Used: K-Means Clustering

K-Means is an unsupervised learning algorithm that partitions data into K clusters.

Steps:

- Select number of clusters (K)
- Initialize centroids randomly
- Assign data points to nearest centroid
- Recalculate centroids
- Repeat until convergence

6. Implementation in WEKA

6.1 Selection of number of clusters (k)

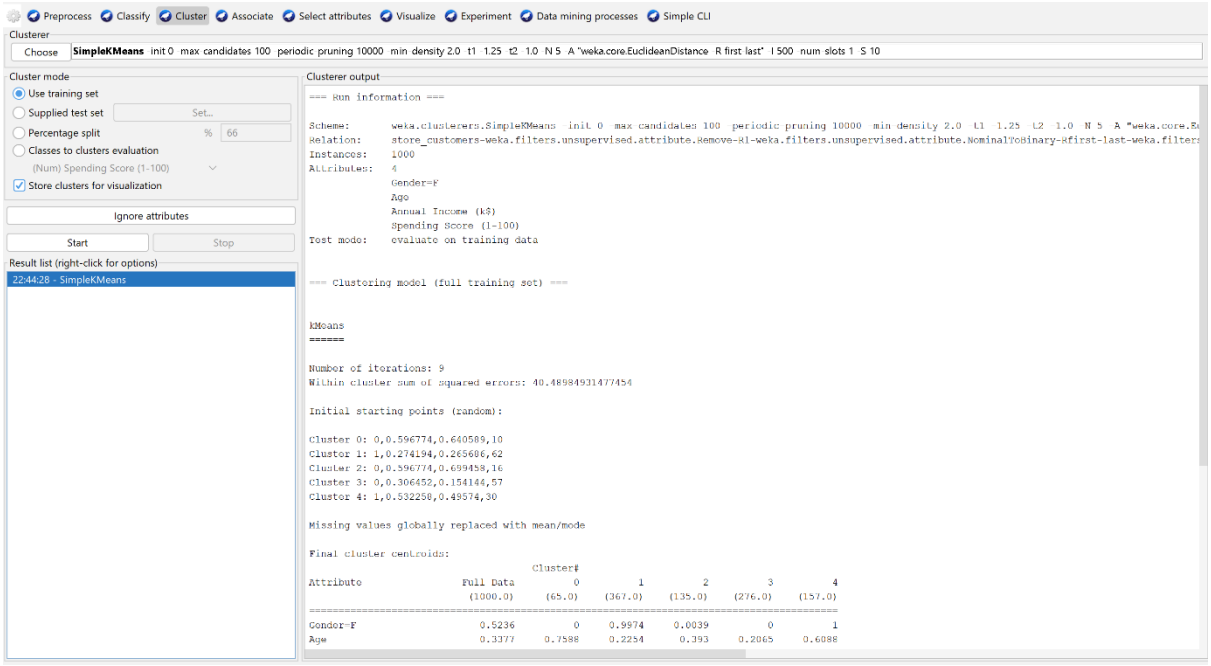
In retail/customer analytics, customers are commonly grouped into **5 meaningful segments** such as:

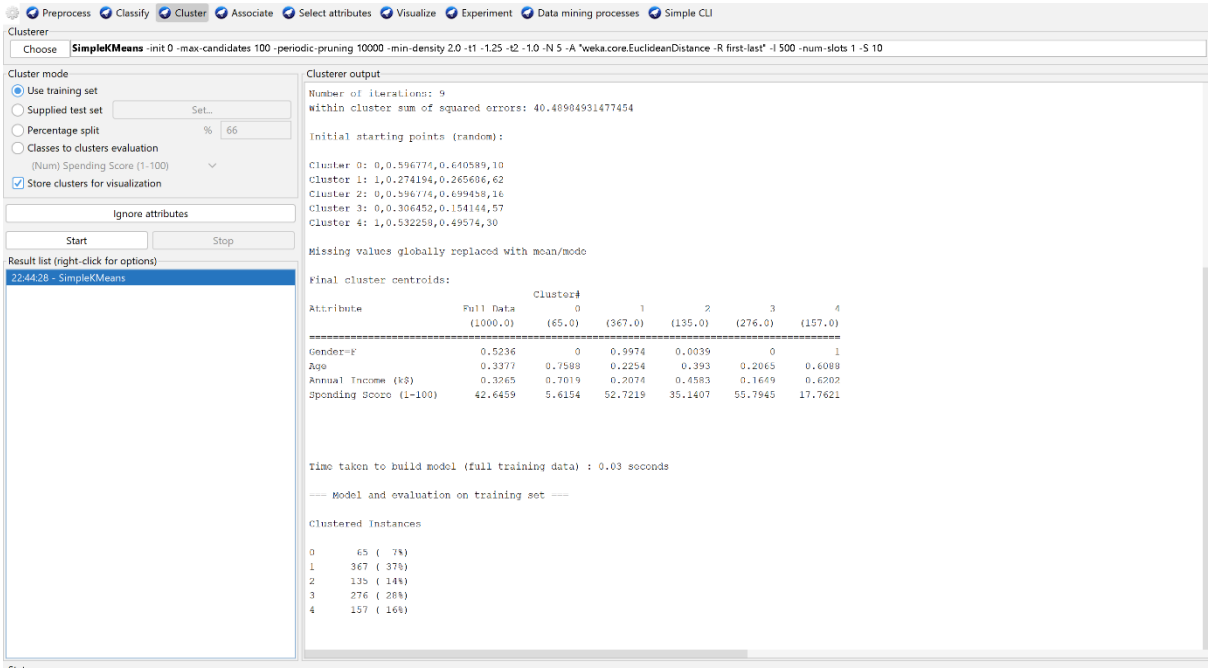
- High Income – High Spending
- High Income – Low Spending
- Low Income – High Spending
- Low Income – Low Spending
- Average Customers

If k is too small, it results in Oversimplified clusters (loss of detail), whereas a greater value of k results in Over-segmentation (confusing results).

Choosing **k = 5** aligns well with real-world business segmentation strategies.

6.2 Selection of algorithm with output screen

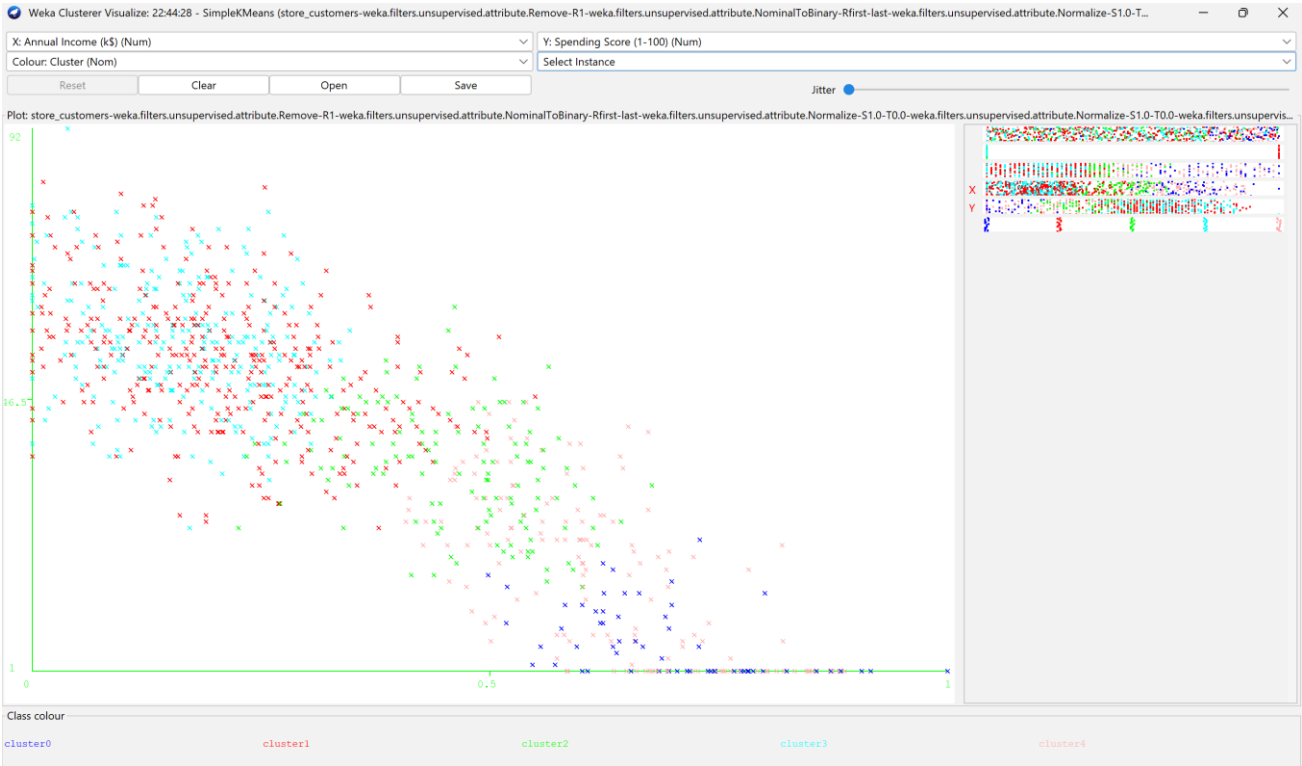




Interpretation:

- Clusters are well distributed
- No empty clusters
- Each group represents a different customer segment
- All clusters have meaningful sizes

6.3 Output (clusters)



1. Overall Pattern (Key Observation)

- The graph shows a negative relationship between:
  - Annual Income (X-axis)
  - Spending Score (Y-axis)

As income increases, spending score tends to decrease for many customers.

Interpretation:

- Not all high-income customers spend more
- Spending behaviour depends on customer type, not just income

2. Presence of 5 Distinct Customer Groups

Each colour represents a cluster → indicating 5 unique customer segments

- Left side → low-income customers

- Right side → high income customers
- Top → high spenders
- Bottom → low spenders

This confirms K = 5 is appropriate

3. Cluster-wise Behavioral Insights

Cluster (Bottom Right) – High Income, Low Spending

- Customers earn more but spend less
- Likely:
    - Conservative buyers
    - Price-sensitive or selective

Insight: **Target with premium offers or personalized recommendations**

Cluster (Middle Left) – Low Income, Moderate Spending

- Limited income but moderate purchases
- Balanced spending behavior

Insight: Offer discounts and loyalty programs

Cluster (Middle Right) – Medium Income, Medium Spending

- Average customers
- Stable purchasing habits

Insight: **Maintain engagement with regular promotions**

Cluster (Upper Left) – Low Income, High Spending

- High spenders despite low income
- Possibly:

- Impulsive buyers
- Trend-driven customers

Insight: **High-value target group → upselling opportunities**

Cluster (Lower Middle/Spread) – Mixed/Transition Group

- Customers between clusters
- Behavior not strongly defined

Insight: **Potential to convert into high-value segments**

4. Density Observations

- Most customers are concentrated in:
  - Low to medium income range
  - Mid-level spending scores

Indicates: **Majority of customers are average spenders**

5. Outliers

- A few points are far from dense regions
- Represent:
  - Unusual spending patterns
  - Rare customer behavior

Useful for:

- Special marketing strategies
- Fraud or anomaly detection (in real-world cases)

6. Business-Level Insights

From this graph, businesses can:

- Identify high-value customers
- Detect low engagement customers
- Design targeted marketing campaigns
- Improve customer retention strategies

7. Conclusion

The scatter plot reveals five distinct customer segments based on income and spending behaviour. While some high-income customers exhibit low spending, certain low-income customers show high spending tendencies. This indicates that spending behaviour is not solely dependent on income, making clustering essential for meaningful customer segmentation.